

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 61



**ECOLOGY
&
ENVIRONMENT-13**



Green signal

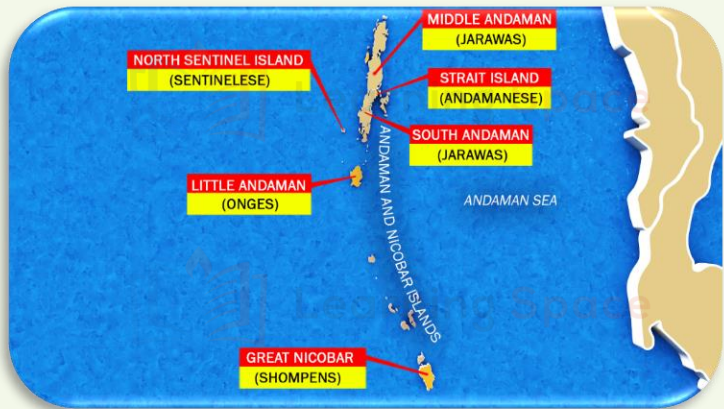
The National Green Tribunal clears ₹92,000 crore Great Nicobar Island mega-infrastructure project, citing its “strategic importance”

- The tribunal finds “no good ground to interfere” with the 2022 environment clearance for the project
- NGT upholds high-powered committee review; dismisses petitions challenging project’s EC scope
- The Environment Ministry directed to protect coral reefs, undertake coral regeneration, and ensure no shoreline erosion or loss of turtle nesting beaches

CLEARANCE GIVEN BY NGT FOR THE GREAT NICOBAR PROJECT (GNP)

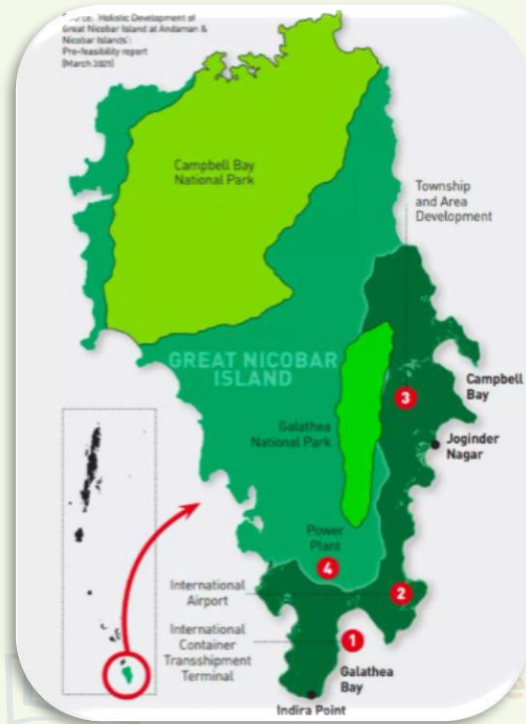
- A bench of the National Green Tribunal (NGT) cleared the way for the Rs. 92,000 crore Great Nicobar Island mega-infrastructure project, noting that “considering the strategic importance” of it and “other relevant considerations”, “we do not find any good ground to interfere”.
- It disposed off applications received related to the Environment Clearance (EC) for the project, with a direction to authorities “to ensure full and strict compliance of EC conditions”.
- The project includes a transshipment port, an airport, a power plant, and a township.
- As per NGT, “a balanced approach is required to be adopted while considering the issue of allowing development of the port on a strategic location and taking adequate steps to carry out the activity strictly in terms of the ICRZ Notification, 2019 instead of prohibiting the activity if the objection is based on apprehension.”

A BRIEF ON GREAT NICOBAR ISLAND



- The island of Great Nicobar is the Southernmost tip of India and a part of the Andaman and Nicobar archipelago that comprises 600-odd islands.
- It is covered with lush rainforests that are sustained by around 3,500 mm of annual rainfall.
- It has an area of 910 sq. km. with mangroves and Pandan forests along its coast.
- It is home to two tribal communities - the Shompen and the Nicobarese.
- The Shompen, around 250 in total, are predominantly hunter-gatherers and are classified as a PVTG within the list of Scheduled Tribes.

MORE ASPECTS ABOUT THE GREAT NICOBAR ISLAND

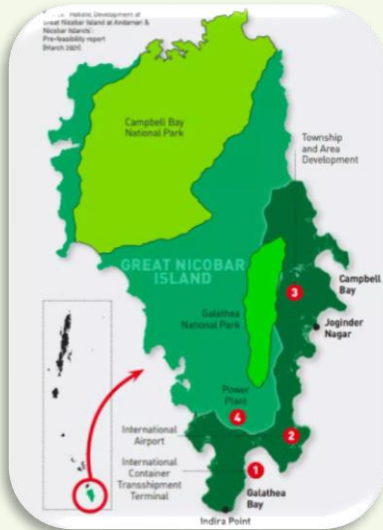


- The Nicobarese community has two groups - the Great Nicobarese and the Little Nicobarese. Campbell Bay predominantly houses Great Nicobarese. They use different dialects of the Nicobarese language.
- The Shompen have their own unique language.
- However, the majority on Great Nicobar Island comprises people who settled on the island from mainland India. Between 1968 and 1975, the Indian Government settled retired military servicemen and their families from various states.

The population of settlers on the island today is around 6,000.

WHAT ARE THE FEATURES OF NITI AAYOG PROJECT?

The idea is that a transshipment port located near one of the world's busiest international sea routes (the Malacca Strait) will allow increased participation in global maritime trade.



- In March 2021, NITI Aayog unveiled a Rs. 72,000 crore plan (then) called “Holistic Development of Great Nicobar Island” at Andaman and Nicobar Islands.
- It includes the construction of an international transshipment terminal, an international airport, a power plant and a township.
- It is to be implemented by Andaman and Nicobar Islands Integrated Development Corporation (ANIIDCO).
- The proposed transshipment port will allow Great Nicobar to participate in the regional and global maritime economy by becoming a major player in cargo transshipment.
- The proposed airport will support the growth of maritime services and enable Great Nicobar Island to attract international and national visitors to experience the outstanding natural environment and participate in sustainable tourism activity.

WHAT ARE THE ENVIRONMENTAL CONCERNS?

1 HUGE ECOLOGICAL COSTS

- It has been heavily criticized for its ecological costs and for potential violations of tribal rights.
- The project requires the diversion of about 130 sq. km. of forest land and the felling of around 10 lakh trees.



2 IMPACT ON WILDLIFE

- In January, 2021, the Indian government denotified two wildlife sanctuaries - the Galathea Bay wildlife sanctuary and the Megapode wildlife sanctuary - to make way for the project.
- At the same time, the Government released a National Marine Turtle Action Plan that lists Galathea Bay as a marine turtle habitat in India.
- The transshipment terminal is expected to be developed at Galathea Bay, one of the world's largest nesting sites for the giant leatherback turtle.
- The Giant Leatherback Turtle and the Nicobar Megapode are listed in Schedule I of the Wildlife (Protection Act), 1972



SUPREME COURT DIRECTIVE TO FSSAI ON FRONT OF PACK WARNING LABELS

- The Supreme Court recently asked the FSSAI to seriously consider introducing front-of-pack warning labels on packaged foods high in sugar, saturated fats, trans fats, and sodium, observing that such labels could advance public health goals.
- It has sought a response within four weeks, effectively nudging the regulator that has debated the issue for years without finalizing the norms.



What do you understand by Ultra-Processed Foods (UPFs)?



- As per UNFAO, UPFs include varieties of sugar (fructose, high fructose corn syrup, fruit juice concentrates, maltodextrin, dextrose, lactose, soluble or insoluble fiber), modified oils (hydrogenated and interesterified oils), hydrolyzed proteins, gluten, when protein and mechanically separated meat.
- To put it in simple words, UPFs contain sugar, salt, fats and oil in high quantities and in carefully calibrated ratios that hit the brain's pleasure centers or bliss points.
- Some examples are - carbonated soft drinks, packaged candies, biscuits, cake mixes, ice cream and chips, artificially sweetened yogurt, pre-prepared meat, cheese, dough and their products, powdered and packaged foods, etc.

Ultra processed foods use additives, stabilizers and preservatives not common in culinary use.

PART OF THE PROBLEM LIES IN REGULATORY AMBIGUITY



- The Central Consumer Protection Authority's guidelines against misleading advertisements and the Food Safety and Standards Act prohibit exaggerated health claims, but they lack clear nutrient-based thresholds to define what constitutes misleading in food marketing.
- An independent study by Label Blind Solutions found that nearly one-third of labeling claims across packaged foods were either non-compliant or they lacked adequate regulatory substantiation.

Simple interpretative warnings such as high in sugar or high in salt are more effective in discouraging unhealthy choices than rating style systems like health star ratings.



In 2023, India committed for 1% SAF landing in international flights by 2027, followed by 5% by 2030, and 15% for domestic flights by 2040.

CERTIFICATION TO IOC TO PRODUCE SUSTAINABLE AVIATION FUEL (SAF)

- In early December 2025, state-owned Indian Oil Corporation (IOC) obtained the country's first certification to produce Sustainable Aviation Fuel (SAF) at its Panipat refinery.
- IOC in August 2025, announced that it would annually produce 35,000 tons of SAF, a group of liquid hydrocarbon fuels that can substitute conventional fuel in aviation.
- SAF may contribute 65% of the emissions reduction needed by aviation to reach net zero carbon dioxide emissions by 2050.
- Certification to IOC is one of the first steps towards India's ambitious targets to reduce aviation emissions.

Routes to sustainability

India's bid to produce sustainable aviation fuel (SAF) relies on availability of feedstock for different pathways. Currently, co-firing of used cooking oil appears to be the most viable route

PRODUCTION PATHWAY	FEEDSTOCK	CURRENT VIABILITY FOR INDIA'S SAF PRODUCTION
Hydroprocessed esters and fatty acids (HEFA): Fats and oils are reacted with hydrogen under high temperature and pressure to remove oxygen and break long molecules into jet-range hydrocarbons. It is a technologically mature pathway and closely resembles processes already used in petroleum refining.	Used cooking oil (UCO)	Some 2.4 million tonnes generated annually, with large potential for HEFA. Collection/traceability issues reduce reliable supply.
	Waste animal fats/tallow	About 142,000-144,000 tonnes of tallow generated annually, primarily for use in soaps/oleochemicals. Only a limited amount is available for SAF.
	Non-edible vegetable oils	Supply chains are not well developed.
Fischer-Tropsch (FT) synthesis: Feedstocks are gasified to produce a mixture of carbon monoxide and hydrogen. This synthesis gas is passed over catalysts to build longer hydrocarbon chains and then upgraded into jet fuel. It is more capital-intensive and less widely deployed than HEFA.	Agricultural residues	Abundant availability, though only a fraction is realistically collectable after competing uses. It could make up around 71% of the estimated 105 million tonnes of biomass available for SAF by 2030.
	Forestry and wood waste	A notable solid biomass source, but dispersed. It could comprise some 16% of biomass for SAF by 2030.
	Municipal solid waste	Available in urban areas but requires segregation and processing infrastructure. It could comprise about 7% of SAF biomass by 2030

DIFFERENT PATHWAYS WITH VARIOUS FEEDSTOCKS

PRODUCTION PATHWAY	FEEDSTOCK	CURRENT VIABILITY FOR INDIA'S SAF PRODUCTION
Alcohol-to-jet (AtJ): Sugar or biomass is fermented into alcohols such as ethanol or isobutanol and then chemically upgraded to form jet-range hydrocarbons. It is still at an early commercial stage.	Sugar/starch sources (such as ethanol)	India is the third-largest ethanol producer, with a feedstock base for AtJ development. Sugar/starch sources may augment as production expands
Power-to-liquid/electro-SAF: Combines captured carbon dioxide with green hydrogen to synthesise liquid hydrocarbons, often via FT chemistry. This route has potential for near-zero lifecycle emissions, but is expensive and limited in scale.	Carbon dioxide and green hydrogen	Currently at a nascent stage

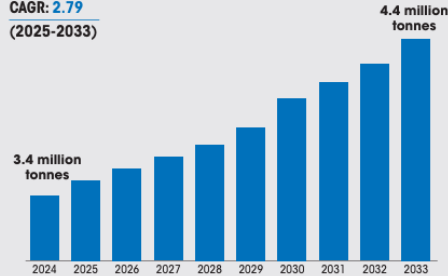
IOC IS FOCUSING ON USED COOKING OIL (UCO)

Growing resource

India generates a vast amount of used cooking oil, which shows high potential for use in sustainable aviation fuel production

CAGR: 2.79

(2025-2033)



Source: "India Biofuel Market Size, Share, Trends and Forecast by Type, Feedstock, and Region, 2025-2033", IMARC Group



- UCO refers to oil or fat left over after food preparation, which is often reused till it turns completely black from burning.
- As per a report by IMARC group, a global management consulting firm, India generated 3.4 million tons of UCO in 2024.
- Once it can no longer be used for cooking, UCO is disposed of in drains or in some cases procured by industries for manufacturing of biodiesel.
- In this case, the UCO is treated with methanol and a catalyst to create fuel.
- UCO is also used to make soaps, chemicals, lubricants, and coatings.

India's theoretical UCO potential is around 2.5 billion liters based on cooking oil consumption, patterns and regulatory limits and reuse.

REPURPOSE USED COOKING OIL (RUCO) WAS LAUNCHED IN 2018



- FSSAI launched a RUCO program in 2018. The double objective of RUCO is to improve public health by preventing reuse of degraded oil and to channel oil into biodiesel production.
- Studies have linked repeated UCO consumption to cancer, heart disease, and liver damage caused by toxic compounds like trans fats formed by reheating oil.
- Improper disposal of UCO pollutes water and soil, clogs drains, and release greenhouse gases.
- Under RUCO, food business operators must avoid reuse of cooking oil more than three times and dispose UCO by giving it to licensed collection agencies, which may be biodiesel producers or parties authorized by them.



NEW COMPREHENSIVE POLICY MANDATES COFIRING OF BIOMASS PELLETS AND MSW DERIVED TORREFIED CHARCOAL

- The Union Ministry of Power has issued a new comprehensive policy mandating the co-firing of biomass pellets and MSW-derived torrefied charcoal in coal-based thermal power plants.
- The policy directs all coal-based thermal power plants in Delhi-National Capital Region (NCR) to use a 5% (by weight) blend of biomass pellets and an additional 2% blend of either biomass pellets or torrefied charcoal made from MSW, along with coal, with effect from financial year 2025-26. Other thermal power plants outside NCR will have to co-fire 5% (by weight) of biomass pellets or torrefied MSW charcoal from 2025-26.
- It specifies that for TPPs located in NCR and adjoining areas, a minimum of 50% of the raw material in pellets must consist of stubble, straw or rice paddy crop residue sourced exclusively from NCR and adjoining regions.

OTHER IMPORTANT ASPECTS OF THIS POLICY



- The torrefaction level of the pellets used maybe based on the OEM recommendation for a particular plant.
- There is also a provision for exemptions on a case-to-case basis to be examined by a committee comprising representatives of the Central Electricity Authority (CEA), CAQM, BHEL, SAMARTH mission, the Union Ministry of Agriculture and Central Power Research Institute.
- The policy also clearly states that surplus agro-residues not being used for animal fodder may be used for pellet production.
- It specifies that for TPPs located in NCR and adjoining areas, a minimum of 50% of the raw material in pellets must consist of stubble, straw or rice paddy crop residue sourced exclusively from NCR and adjoining regions.
- For TPPs, under Section 62 of the Electricity Act, 2003, the cost of co-firing biomass pellets and MSW charcoal shall be passed through the Energy Charge Rate (ECR). If the project has a fuel cost pass through clause in its power purchase agreement, the added cost shall be passed through the ECR.



- Cofiring MSW derived charcoal along with coal will lead to several operational, technical and environmental challenges.
- MSW charcoal typically contains high chlorine, alkali metals and heterogeneous components which can cause increased slagging, fouling and accelerated corrosion of boiler pressure parts, air preheaters and ducting, as per analysts. The fuel's variable calorific value and higher moisture content may result in unstable combustion, reduced boiler efficiency, and higher unburned carbon.
- The fine and low-density ash associated with MSW fuels can overload the electrostatic precipitator reduce particulate capture efficiency and adversely affect downstream systems such as flue gas desulfurization.
- Cofiring may also raise emissions of dioxins, furans, and heavy metals and acidic gases.
- Handling and storage challenges, including dustiness and fire hazards further increase operational complexity.

MSW HAS ALSO BEEN THE MAIN FUEL FOR ALL WASTE TO ENERGY (WTE) PLANTS IN INDIA



- WTE plants face multiple challenges and concerns over health and environmental impacts due to emissions, proximity to cities, process inefficiency stemming from feedstock quality and issues of overall economic feasibility.
- India's waste is unsegregated with low calorific value, which burns inefficiently, creates toxic emissions and requires subsidies just to stay operational.
- Using the same substandard fuel for torrefaction to produce the so-called biocoal (torrefied charcoal) doesn't solve the problem, it simply shifts it.
- Torrefaction removes moisture and increases energy density, but it does not remove contaminants, it only relocates them.

ENVIRONMENTAL HAZARDS WILL PERSIST WITH THE PRESENT DECISION!



- The chlorine present in plastics and food waste is released as corrosive hydrogen chloride gas, creating acidic liquid waste and attacking plant equipment.
- The non-volatile heavy metals such as lead and copper from e-waste batteries or paints will become more concentrated in the final biocoal essentially priming it with catalysts for dioxin formation during co-firing.
- The fundamental lesson remains unchanged - you cannot fix a contaminated feedstock completely, merely by changing the process and technology!

The real solution lies at the source with proper segregation.

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